

Catalog Service Design Document

Purpose: Source of truth for Movie, Theatre, Screen and Show metadata.

Responsibilities

Manage movies, theatres, screens and schedule shows. Publish MovieCreated, MovieUpdated, MovieDeleted, ScreenCreated, ScreenUpdated, ScreenDeleted, ShowCreated, ShowUpdated and ShowCancelled events.

Owned Entities

Movie, Theatre, Screen, Show

Not Responsible For

Seat availability, booking lifecycle, payments and notifications.

Business Rules

- Screen belongs to one Theatre.
- Show belongs to one Movie and one Screen.
- Show end time must be after start time.
- No overlapping shows on the same screen.
- Soft delete for Theatre/Screen/Movie where appropriate.

Database Design (catalog_db)

Movie: movie_id, title, description, duration_minutes, language, genre, release_date, created_at, updated_at, version, is_deleted

Theatre: theatre_id, name, address, city, state, country, created_at, updated_at, version, is_deleted

Screen: screen_id, theatre_id, name, screen_type, created_at, updated_at, version, is_deleted

Show: show_id, movie_id, screen_id, start_time, end_time, status, created_at, updated_at, version, is_deleted

Indexes

Movie(title), Movie(release_date), Theatre(city), Theatre(name,city), Screen(theatre_id), Show(movie_id), Show(screen_id,start_time), Show(start_time)

Architecture Decision

For Version 1 of the project, Seat ownership will remain in the Inventory Service so development effort stays focused on concurrency, locking, Saga, Kafka, and distributed systems. Catalog owns only Movie, Theatre, Screen and Show metadata.